

- 3 (a) (i) Kinetic energy of one gas particle (atom / molecule) = $\frac{1}{2}mc_{rms}^2 = \frac{3}{2}kT$
 where m is the mass of one gas particle/atom/molecule. B1

For one mole of gas containing N_A particles, the total kinetic energy is given by:

$$\frac{1}{2}N_A mc_{rms}^2 = \frac{3}{2}N_A kT \text{ ----- (1)}$$

From the equation of state of an ideal gas for 1 mole of ideal gas:
 $pV = (1)RT = N_A kT$

Substituting $N_A kT = RT$ into (1) gives: $\frac{1}{2}N_A mc_{rms}^2 = \frac{3}{2}RT$

Simplifying to get: $c_{rms} = \sqrt{\frac{3RT}{N_A m}} = \sqrt{\frac{3RT}{M}}$

Where $N_A m$ is the mass of 6.02×10^{23} particles = molar mass M .

(ii) $c_{r.m.s.} \propto \frac{1}{\sqrt{M}} \quad \frac{c_{r.m.s. \text{ of oxygen molecules}}}{c_{r.m.s. \text{ of nitrogen molecules}}} = \sqrt{\frac{28}{32}} = 0.935 \text{ or } 0.94$
 A1

(b) (i) $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle = \frac{1}{3} \rho \langle c^2 \rangle = \frac{1}{3} \times 1.50 \times 10^5 \times (4.85 \times 10^5)^2$ M1

$= 1.18 \times 10^{16} \text{ Pa}$ A1

(ii) Forces between nuclei/particles are not negligible. (ignore "attractive") M1

Forces are repulsive (at that density) contributing to an increase in pressure A1

OR

Volume of particles is not negligible (at that density) M1

Resulting in a higher rate (or frequency) of collision compared to the expected rate, causing the actual pressure to be higher than the expected value in (b)(i). A1

(c)

Process	w / kJ	q / kJ	⊗U / kJ
A to B	<u>19.2</u>	67.2	<u>- 48.0 [B1]</u>
B to C	0	<u>48.0</u>	<u>48.0 [B1]</u>
C to A	31.6	31.6	0

